Inorganic Cumulative Exam
9-26-2025

Answer any 5 of the following 10 questions (20 points each). Any answers provided beyond the requested five will not be graded. Remember: Your answers to these questions are meant to provide evidence of broad, graduate-school level chemistry knowledge. Consider this in your choice of terminology and examples as answers to these questions.

Molecular Inorganic Chemistry

1) Metal oxo complexes can be utilized in a variety of redox reactions. What are the oxidation state and d electron count of the metal in the following compounds?
A) $(acac)_2MoO_2$, B) $(acac)_2FeO$, C) Cp_2VO
(acac = acetylacetonate, Cp = cyclopentadienyl).

2) Predict the reaction between trimethyl phosphine and the compounds in question 1
A) $(acac)_2MoO_2$, B) $(acac)_2FeO$, C) Cp_2VO

3) Define the "oxo wall" and describe what it tells us about metal oxo compounds?

4) Propose a strategy to "break the oxo wall" and provide an example if available?

5) Describe three different reaction mechanisms that are used to reduce metal oxo complexes.

Solid State Inorganic / Materials Chemistry

6) Propose two methods to synthesize a metal oxide surface that is under 100 nm thick.

7) Provide two methods for characterizing the metal oxide surface you generated in question 6.

8) Describe the processes of annealing. What is it and what are the consequences of annealing a metal oxide surface?

9) The Mars–van Krevelen mechanism is currently the accepted mechanism for the oxidation of organic molecules by oxygen catalyzed at metal oxide surfaces. Briefly summarize this mechanism.

10) Alivisatos and co-workers described cationic replacement reactions in nanocrystals in the 1990s/early 2000s. Predict the change in one physical property when you replace Cd atoms in a CdSe nanocrystal with Ag. Briefly explain your answer.